

IEEE Signal Processing Cup 2026

AV Zoom: Real-Time Audio-Visual Zooming on Smartphones

Orientation & Training session | 21 November 2025

IEEE SP Cup 2026 Organizer:

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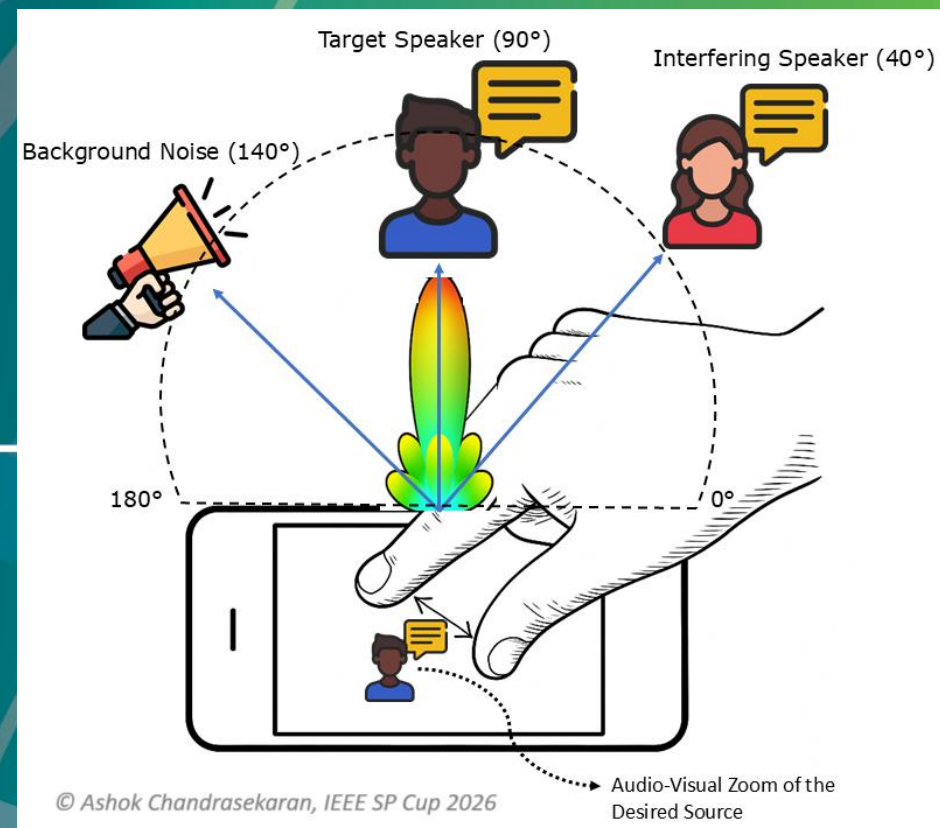
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Welcome & Overview of IEEE SP Cup 2026



Flagship Student Competition of IEEE SPS (held at ICASSP)

“This challenge aims to inspire young minds to tackle practical technical problem”.

- Organized by the IEEE Signal Processing Society
- SP Cup Challenge: AV Zoom - Real-Time Audio-Visual Zooming on Smartphones
- Overview of tasks, phases, evaluation criteria, deadlines, and submission guidelines

Prizes for Finalists:

 **1st Place:** \$5,000

 **2nd Place:** \$2,500

 **3rd Place:** \$1,500

Travel support for up to 3 students

Complimentary conference registration

Access to SPS Job Fair & Banquet (subject to availability)



What is Audio-Visual Zooming?

*Achieving synchronization between **what is seen** and **what is heard** is crucial for enhancing user experience.*

- Synchronizes camera zoom with audio focus
- Enhances the clarity of the desired target (speaker's) audio
- Suppresses interfering audio
- Useful for smartphone-based applications such as vlogging, meetings, and more



Source: LG G2 - Audio Zoom, 2013

“Audio Focus technology enables the selection of the desired direction and attenuates incoming sounds from other directions. When the strength of this effect is controlled dynamically—for example, together with the video camera’s visual zooming—the effect is called Audio Zoom.”

— Nokia’s Audio Zoom description (OZO Audio brochure, Nokia Technologies © 2022)

Motivation

- Smartphones have a limited number of microphones → low spatial resolution
- Audio-visual zooming remains unreliable on many devices
- Need for low-power, real-time, on-device solutions

The proposed system should integrate the following components:

- Real-time audio zooming using the microphone array to focus on specific sound sources.
- Visual alignment with the target sound source, ensuring that what is heard matches what is seen.

All components should be optimized for edge devices—specifically smartphones—with emphasis on low power, low latency, and fully on-device operation.

Competition Overview



Competition Phases:

Phase 1: Proof of Concept (PoC) submission

Phase 2: MATLAB Simulations (Tasks 1 & 2) — video presentation and report submission — technical evaluation

Phase 3: Real-Time Mobile App Demonstration — video presentation and report submission — technical evaluation

Note:

- Each phase has individual scoring
- Top 3 teams advance to the Finals at ICASSP 2026
- *After submitting the Phase 1 Proof of Concept, teams are permitted to revise or improve their methodology and use the updated approach in both Phase 2 and Phase 3*



Task Description

Signal Model:

The n -th snapshot of the received signal at the m -th microphone is expressed as

$$y_m(n) = s(n) + i_m(n) + v_m(n), n = 0, 1, \dots, N_s - 1$$

where

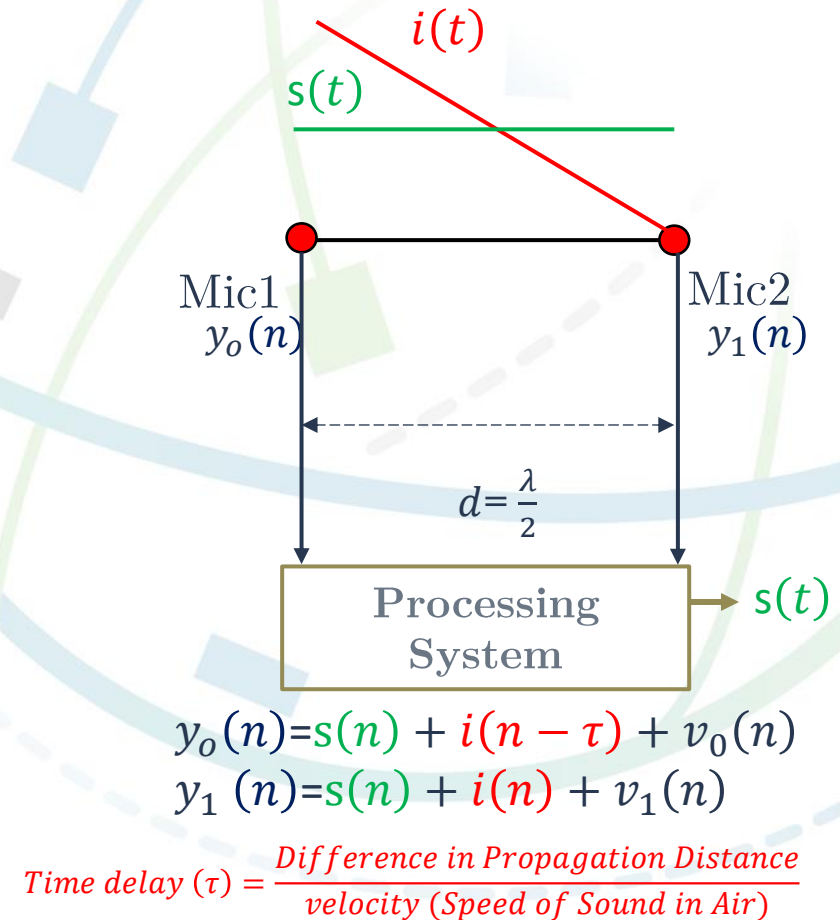
$s(n)$ is the target signal to be zoomed in,

$i_m(n)$ represents the total received interference at the m -th microphone

$v_m(n)$ denotes the additive noise.

N_s is the total number of time samples.

Note: Since the target signal is incident from broadside, it arrives with identical delay at both microphones.



Task Description

Task 1: Anechoic Simulation (Ideal Case)

Simulate an ideal anechoic scenario using MATLAB (toolboxes) for room acoustics and array signal processing.

Specifications:

1. Room Setup

- Type: Anechoic chamber (no reflections, no reverberation)
- Absorption: Perfect (no reflections)
- Purpose: Baseline testing (ideal separation)
- Dimensions: 16 ft $\times$ 16 ft $\times$ 16 ft ($\approx 4.9 \times 4.9 \times 4.9$ m)
- Reverberation Time (RT60): ≈ 0.0 s
- Simulation in MATLAB (any one of the following):
 - Image Source Method (ISM):
Use `acousticRoomResponse` function, or
Use `rirGenerator` with `ReflectionOrder = 0`, or
Use `audioRoom('Method','imagesource')` with `Reflectivity = 0` (direct path only).
 - Stochastic Ray Tracing (SRT):
Use `audioRoom('Method','stochastic')` with `Reflectivity = 0` to simulate anechoic direct path.

Task Description

Task 1: Anechoic Simulation (Ideal Case)

2. Microphone Array

- Position: At room center
- Mic 1: (2.41, 2.45, 1.5) m
- Mic 2: (2.49, 2.45, 1.5) m
- Array center: (2.45, 2.45, 1.5) m
- Configuration: 2-microphone linear array
- Spacing (d): 8 cm
- Sampling Rate: 16 kHz



Task Description

Task 1: Anechoic Simulation (Ideal Case)

3. Sources

- Source 1 (Target Speaker – Male Speech)
- Azimuth: 90° (directly in front of array)
- Height: 1.5 m
- Position: (2.45, 3.45, 1.5) m
- Dataset Requirement: Participants are expected to use the male speech file from MUSAN & LibriSpeech corpus:

E.g.

speech-librivox-0003.wav (Male speech – MUSAN)

speech-librivox-0000.wav (Male speech – MUSAN)

This file can be accessed through the dataset links provided below:

<https://www.openslr.org/12>

<https://www.openslr.org/17>

Task Description

Task 1: Anechoic Simulation (Ideal Case)

3. Sources

- Source 2 (Interference)
- Azimuth: 40° (to the right-front)
- Height: 1.5 m
- Position: (3.22, 3.06, 1.5) m
- Dataset Requirement: Participants are expected to use the female speech file from the MUSAN & LibriSpeech corpus:

E.g.

1462-170138-0000.flac (Female speech -LibriSpeech)

speech-librivox-0100.wav (Female speech -MUSAN)

This file can be accessed through the dataset links provided below:

<https://www.openslr.org/12>

<https://www.openslr.org/17>

Task Description

Task 1: Anechoic Simulation (Ideal Case)

4. Mixture Conditions

- Signal-to-Interference Ratio (SIR): 0 dB (equal power target & interferer)
- Signal-to-Noise Ratio (SNR): 5 dB (additive sensor (microphone) noise)
- Sensor (microphone) Noise type: White Gaussian noise (for controlled simulation)

Task Description

Task 1: Anechoic Simulation (Ideal Case)

5. Evaluation metrics

- Evaluation of the proposed solution for the simulation setup should be performed using the following metrics:
- Scale-Invariant Signal-to-Distortion Ratio (SI-SDR)
- Perceptual Evaluation of Speech Quality (PESQ) (ViSQOL — MATLAB's updated objective speech quality metric)
- Short-Time Objective Intelligibility (STOI)

Task Description

Task 2: Reverberant Room Simulation (Realistic Case)

1. Room Setup

- Type: Reverberant room
- Dimensions: $\approx 4.9 \times 4.9 \times 4.9$ m (same as Task 1)
- Reverberation Time (RT60): ≈ 0.5 s (moderate reverberation)
- Simulation in MATLAB (any one of the following):
 - Image Source Method (ISM):
Use `acousticRoomResponse` function, or
Use `rirGenerator` with `ReflectionOrder > 0`, or
Use `audioRoom('Method','imagesource')` with reflectivity values set to achieve $RT60 \approx 0.5$ s.
 - Stochastic Ray Tracing (SRT):
Use `audioRoom('Method','stochastic')` with appropriate absorption coefficients to yield $RT60 \approx 0.5$ s.
- Absorption Coefficients: Walls, ceiling, floor set to yield $RT60 \approx 0.5$ s

Task Description

Task 2: Reverberant Room Simulation (Realistic Case)

Hint :

RT60 = 0.5 s

Room Impulse Response (RIR) using the image source method

Steps:

- Compute room volume & surface area
- Compute absorption coefficient from RT60
- Compute reflection coefficient
- Generate RIR using image source method for direct & reflection
- Sanity check: Measure actual RT60 by using Schroeder backward integration
- Apply RIR to source signals

Result: microphones receive reverberant signals consistent with $RT60 = 0.5$ s.

Task Description

Task 2: Reverberant Room Simulation (Realistic Case)

2. Microphone Array

Same as Task 1 (2-element linear array, 8 cm spacing, 16 kHz sampling)

3. Sources

Same as Task 1

Source 1: Male speech (target, at 90°)

Source 2: Female speech (interferer, at 40°)

4. Mixture Conditions

SIR: 0 dB

SNR: 5 dB

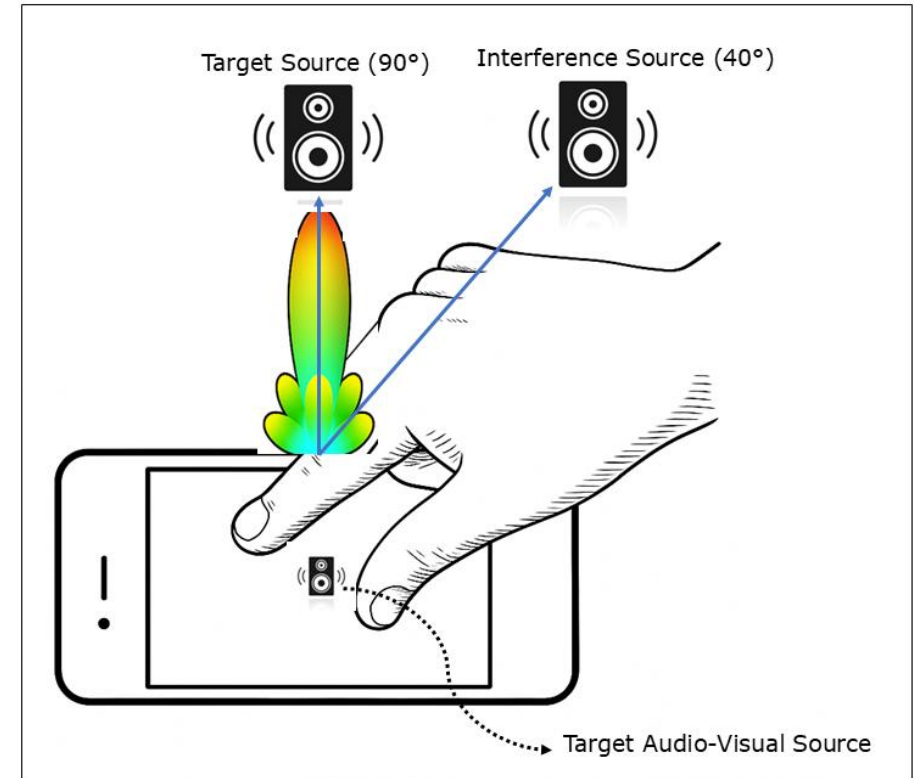
5. Evaluation Metrics

Same as Task 1: SI-SDR, PESQ (ViSQOL — MATLAB's updated objective speech quality metric) and (STOI)

Task Description

Task 3: Mobile Application Integration and Real-Time Evaluation

- To make this task manageable within the competition timeline, the final real-time demonstration can be conducted in a controlled environment, as illustrated in Figure 2 (official SP cup document).
- Develop a mobile application (Android or iOS) that integrates the proposed solution for a 2-element microphone array and 2 sources.
- Evaluation setup:
 - The same two sources used in Task 1 and Task 2 will be played back through loudspeakers.
 - The speakers should be placed with the same angular spacing specified in Task 1 and Task 2.
 - The evaluation must be conducted at the same SIR and SNR levels as defined earlier.
- Quantitative Evaluation Metrics
 - Audio quality (SI-SDR, PESQ (ViSQOL), STOI)
 - System efficiency (latency, memory)



Useful References



- A. M. Elbir, K. V. Mishra, S. A. Vorobyov, and R. W. Heath, “Twenty-five years of advances in beamforming: From convex and nonconvex optimization to learning techniques,” *IEEE Signal Processing Magazine*, vol. 40, no. 4, pp. 118–131, June 2023.
- A. A. Nair, A. Reiter, C. Zheng, and S. Nayar, “Audiovisual zooming: What you see is what you hear,” *Proceedings of the 27th ACM International Conference on Multimedia (ACMMM)*, pp. 1107–1118, 2019.
- A. Khandelwal, E. Goud, Y. Chand, S. Prasad, N. Agarwala, and R. Singh, “Two-channel audio zooming system for smartphone,” *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2021.
- A. Khandelwal, *Two-Channel Robust Audio Zooming System for Smartphone*, Master’s Thesis, June 2018.
- N. Q. K. Duong, P. Berthet, S. Zabre, M. Kerdranvat, A. Ozerov, and L. Chevallier, “Audio zoom for smartphones based on multiple adaptive beamformers,” *International Conference on Latent Variable Analysis and Signal Separation*. Springer, pp. 121–130, 2017.
- Yu, Meng and Dong Yu. “Deep Audio Zooming: Beamwidth-Controllable Neural Beamformer.” Tencent AI Lab, 2023.



Evaluation Protocols



The evaluation of the competition will be conducted across three phases, each with specific procedures for assessing performance.

Phase 1 – Proof of Concept (PoC)

- Remote evaluation of submitted reports.
- Judges assess innovation, technical feasibility, and clarity of documentation.
- Teams are ranked using weighted scores (Innovation 20%, Technical Feasibility 20%, Documentation 10%).
- *Teams will be ranked based on total Phase 1 scores out of 50. The top teams will advance to **Phase 2** for technical evaluation (Task 1 and Task 2).*



Evaluation Protocols

Phase 2 – Video Presentation (VP)

- Remote evaluation of MATLAB simulation results.
 - Task 1:** Anechoic environment (ideal)
 - Task 2:** Reverberant environment ($RT60 \approx 0.5$ s)
- Signal conditions: $SIR = 0$ dB, $SNR = 5$ dB
- Teams run simulations, record quantitative metrics (SI-SDR, PESQ (ViSQOL), STOI), and submit a video demonstrating methodology and results.
- Teams are ranked based on measured metric values.

Evaluation Protocols

Phase 3 – Real-Time Demonstration (RTD)

- Live evaluation on a smartphone in a controlled environment.
- Two audio sources played at specified angles; participants demonstrate real-time audio-visual zooming with a pinch gesture.
- Quantitative metrics recorded: SI-SDR, PESQ (ViSQOL), STOI, latency, memory usage
- Qualitative assessment: innovation, usability, report and presentation clarity
- Judges combine quantitative and qualitative performance for final rankings.

Submission Structure

Folder Contents (for Tasks 1 and 2)

Each `.mat` file must contain the following variables:

`target_signal` – Clean waveform of the male target speaker

`interference_signal` – Waveform of the interference signal

`mixture_signal` – Combined mixture audio at specified SNR/SIR

`rir_data` – Room Impulse Response (anechoic or reverberant, as applicable)

`processed_signal` – Output of the participant's audio-visual zooming algorithm

`metrics` – Evaluation results: OSINR, PESQ, STOI

`params` – Metadata including microphone positions, array spacing, sampling rate, source azimuth/height,

SNR/SIR values

Directory Structure

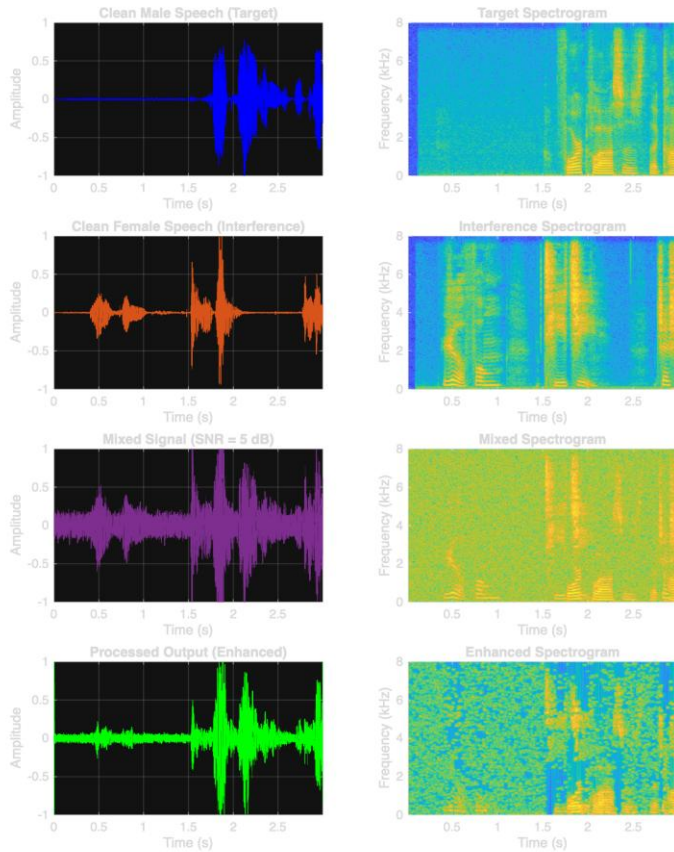
/Task1_Anechoic/

/Task2_Reverberant/

/Task3_RealTime/

Sample Simulation Results

Task 1: Anechoic Room Beamforming Results



EVALUATION METRICS

Perceptual Quality:

STOI:

VISQOL:

Distortion:

S1-SDR (Output):

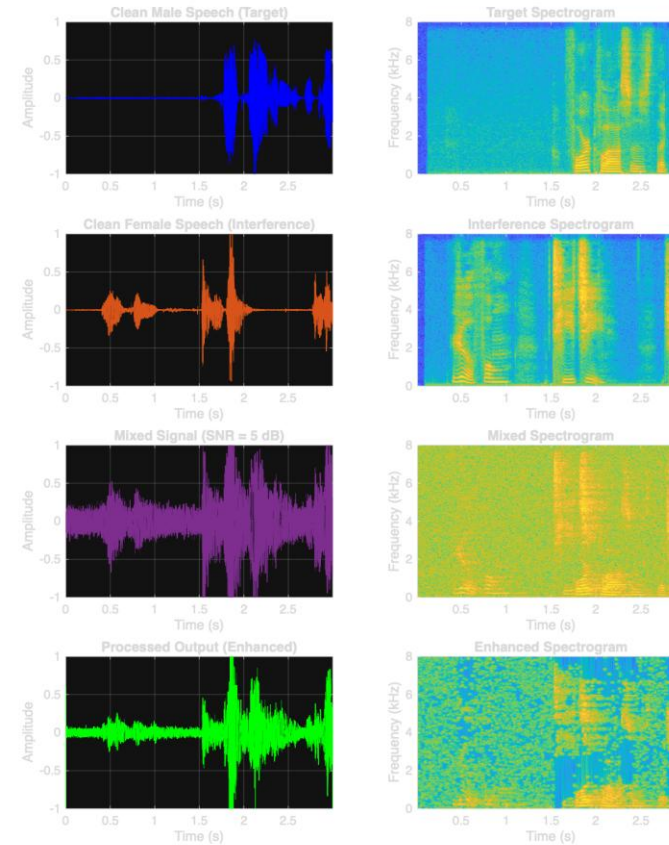
Configuration:

SIR:

SR:

STSD:

Task 2: Reverberant Room Beamforming Results



EVALUATION METRICS

Perceptual Quality:

STOI:

VISQOL:

Distortion:

S1-SDR (Output):

Configuration:

SIR:

SR:

STSD:

Important Dates



Competition Announcement	20 October 2025
Team Registration Deadline	10 November 2025
Phase 1 Team Work Submission Deadline	11 December 2025
Announcement of the Phase 1 Results	21 December 2025
Phase 2 Team Work Submission Deadline	08 February 2026
Announcement of 3 Finalists Teams	02 March 2026
Final Round of Competition at ICASSP 2026	4 May 2026

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**Thank You
All the best!**

*We're excited to support your journey and we look forward to seeing the innovative work you will create for the **IEEE Signal Processing Cup 2026**.*